ENTERPRISE NETWORK SECURITY IMPLEMENTATION

Fortinet Capstone Project — Production Engineering Documentation

Author: Amr | Platform: FortiOS 7.0 & Cisco IOS | Topology: Multi-Site HQ, BR1, BR2 & SSL-VPN

1. Executive Summary & Architecture Overview

This capstone engineering document details the implementation of an end-to-end secure enterprise network designed in accordance with enterprise security benchmarks and Zero-Trust principles. The topology simulates a complete corporate enterprise comprising:

- **Corporate Headquarters (HQ):** High Availability (Active-Passive) FortiGate cluster, redundant WAN links, Core-HQ 802.1Q distribution switch, and dedicated access switches delivering 5 micro-segmented VLAN tiers.
- **Branch 1 Site (BR1):** Next-Generation FortiGate perimeter with local user and server VLAN segmentation.
- **Branch 2 Site (BR2):** Redundant branch perimeter providing local segmentation and web application hosting.
- **Remote Teleworkers:** SSL-VPN client-to-site gateway configured on HQ WAN for secure remote administrative and corporate access.
- **Simulated Provider Core (ISP):** Cisco IOS multi-hop routing fabric interconnecting sites across independent public /30 point-to-point subnets over Cloud0.

2. Master IP Addressing & VLAN Matrix

2.1 Management & WAN Links Matrix

Device	Interface	IP Address	Subnet Mask	Gateway	Role / Description
ISP-HQ	Gi0/0	192.168.85.201	255.255.255.0	192.168.85.2	Cloud0 External Management Gateway
ISP-HQ	Gi0/1	203.0.113.1	255.255.255.252	—	Point-to-point link to WAN-HQ
WAN-HQ	Gi0/1	203.0.113.2	255.255.255.252	203.0.113.1	Uplink to ISP-HQ
WAN-HQ	Gi0/0	203.0.113.5	255.255.255.252	—	Downlink to HQ WAN Switch / Firewalls
Main-HQ (HA)	port1	203.0.113.6	255.255.255.252	203.0.113.5	HQ Public Cluster WAN Gateway
ISP-BR1	Gi0/0	192.168.85.202	255.255.255.0	192.168.85.2	Branch 1 ISP Cloud Interface
ISP-BR1	Gi0/1	203.0.113.9	255.255.255.252	—	Link to WAN-BR1
WAN-BR1	Gi0/1	203.0.113.10	255.255.255.252	203.0.113.9	Uplink to ISP-BR1
WAN-BR1	Gi0/0	203.0.113.13	255.255.255.252	—	Downlink to FGT-Branch1
FGT-Branch1	port1	203.0.113.14	255.255.255.252	203.0.113.13	Branch 1 Public WAN Interface
ISP-BR2	Gi0/0	192.168.85.203	255.255.255.0	192.168.85.2	Branch 2 ISP Cloud Interface
ISP-BR2	Gi0/1	203.0.113.17	255.255.255.252	—	Link to WAN-BR2

WAN-BR2	Gi0/1	203.0.113.18	255.255.255.252	203.0.113.17	Uplink to ISP-BR2
WAN-BR2	Gi0/0	203.0.113.21	255.255.255.252	—	Downlink to FGT-Branch2
FGT-Branch2	port1	203.0.113.22	255.255.255.252	203.0.113.21	Branch 2 Public WAN Interface

2.2 Corporate LAN Micro-Segmentation Matrix

Site	VLAN ID	Segment Name	Subnet	Gateway IP	DHCP Scope / Static Hosts	Parent Port
HQ	10	HQ-Users-VLAN10	192.168.10.0/24	192.168.10.1	192.168.10.100 - .200 (DHCP)	port2.10
HQ	20	HQServersVLAN20	192.168.20.0/24	192.168.20.1	192.168.20.20 (Web), .10 (VPC11)	port2.20
HQ	25	HQServersVLAN25	192.168.25.0/24	192.168.25.1	192.168.25.10 (API Tier)	port2.25
HQ	27	HQServersVLAN27	192.168.27.0/24	192.168.27.1	192.168.27.10 (Database Tier)	port2.27
HQ	30	HQManagVLAN30	192.168.30.0/24	192.168.30.1	192.168.30.10 (Admin PC)	port2.30
BR1	10	VLAN10-Users	10.10.10.0/24	10.10.10.1	10.10.10.10 - .50 (Static/DHCP)	port2.10
BR1	20	VLAN20-Servers	10.10.20.0/24	10.10.20.1	10.10.20.10 (Internal Server)	port2.20
BR2	10	VLAN10-Users	10.30.10.0/24	10.30.10.1	10.30.10.10 - .50 (Static/DHCP)	port2.10
BR2	20	VLAN20-Servers	10.30.20.0/24	10.30.20.1	10.30.20.10 (Web Server)	port2.20

3. High Availability (HA Active-Passive) Design

Headquarters perimeter resilience is ensured by an Active-Passive FortiGate cluster:

- **Nodes:** Main-HQ (Priority 200, Active Master) and Backup-HQ (Priority 100, Standby Slave).
- **Heartbeat Interfaces:** Redundant dual links over `port3` and `port4` (weight 50 each).
- **State Synchronization:** Real-time session pickup preserves TCP connections during failover.
- **Virtual MAC (VMAC):** Shared cluster MAC address eliminates ARP cache delays on WAN and Core switches.
- **Flapping Prevention:** `override disable` ensures stability by preventing preemption until planned maintenance.

```
# FortiGate HA Configuration Applied
config system ha
    set group-name "HQ-HA-Cluster"
    set mode a-p
    set hbdev "port3" 50 "port4" 50
    set session-pickup enable
    set override disable
    set priority 200
end
```

4. Site-to-Site IPsec VPN & Hub-and-Spoke Routing

The corporate WAN utilizes route-based IPsec tunnels terminating at Headquarters:

- **Tunnels:** `To-BR1` (HQ ↔ Branch 1) and `To-BR2` (HQ ↔ Branch 2).
- **Hub-and-Spoke Transit Routing:** Branch 1 (10.10.0.0/16) and Branch 2 (10.30.0.0/16) communicate

through HQ without requiring direct mesh links. HQ inspects and routes transit traffic via firewall policies `vpn_To-BR1_local_0` and `BR2-to-HQ`.

- **Proposals:** DES/AES encryption, SHA hashing, Diffie-Hellman Group 14/5, PSK authentication.

5. Firewall Policy Matrix & Zero-Trust Micro-Segmentation

ID	Policy Name	From → To	Source → Destination	Service	NAT	UTM Inspection Profiles
2	HQ-Users-to-Internet	VLAN10 → port1	HQ-Users-Subnet → all	DNS, HTTP, HTTPS, PING	ENABLE	Antivirus, IPS, WebFilter, AppControl, SSL-Insp
3	WAN-to-HQ-Web-Server	port1 → VLAN20	all → HQ-Web-VIP-8080	ALL	OFF	Destination NAT VIP (8080 → 80)
4	SSLVPN-to-HQ-Web-Server	ssl.root → VLAN20	all → HQ-Web-Server	HTTP, PING	OFF	Antivirus, IPS, WebFilter, AppControl, SSL-Insp
5	vpn_To-BR1_local_0	VLAN10 ↔ To-BR1	To-BR1_local ↔ To-BR1_remote	ALL	OFF	Branch 1 User IPsec VPN Policy
6	vpn_To-BR1_local_1	VLAN20 ↔ To-BR1	To-BR1_local ↔ To-BR1_remote	ALL	OFF	Branch 1 Server IPsec VPN Policy
9	InterVLAN-All	VLAN10/20/30 ↔ All	all → all	ALL	OFF	Internal Routed Communications
10	SRV-API	VLAN20 → VLAN25	all → all	ALL	OFF	Web-to-API Layer Micro-segmentation
11	DB-API	VLAN27 → VLAN25	all → all	ALL	OFF	Database-to-API Layer Micro-segmentation
12	HQ-to-BR2	VLAN10 ↔ To-BR2	To-BR1,2_local ↔ To-BR2-Remote	ALL	OFF	Branch 2 User IPsec VPN Policy
13	BR2-to-HQ	VLAN20 ↔ To-BR2	To-BR1,2_local ↔ To-BR2-Remote	ALL	OFF	Branch 2 Server IPsec VPN Policy

6. NAT & Virtual IP (VIP) Server Publishing

Location	VIP Name	External IP / Port	Internal IP / Port	Function
Headquarters	HQ-Web-VIP-8080	203.0.113.6:8080	192.168.20.20:80	Port forwarding public traffic to HQ Web Server
Branch 1	VIP_Internal_Server	203.0.113.10:8080	10.10.20.10:80	Publishing internal Branch 1 Server to WAN
Branch 2	VIP-B2-Web-Server	203.0.113.22:8080	10.30.20.10:80	Publishing internal Branch 2 Web Server to WAN

7. Remote Access SSL-VPN Implementation

Secure client-to-site connectivity for teleworkers and off-site administrators:

- **Listening Endpoint:** FortiGate-HQ port1 (`https://203.0.113.6:443`).
- **Tunnel Address Pool:** `SSLVPN-Tunnel-Range` (10.212.134.200 - 10.212.134.210).
- **Portal Mode:** `full-access` (Tunnel Mode with full gateway enforcement).

- **Access Control:** Policy 4 restricts SSL-VPN clients exclusively to `HQ-Web-Server` (192.168.20.20) with active deep UTM inspection.

8. Engineering Troubleshooting History & Root-Cause Analysis

- **Cloud0 Duplicate Address Conflict:** Resolved `%IP-4-DUPADDR` conflict between virtual routers and VMnet8 by assigning dedicated static IPs (HQ: .201, BR1: .202, BR2: .203) and routing through default gateway `192.168.85.2`.
- **HA Inode Glitch & Maintainer Recovery:** Recovered disk corruption (`EXT3-fs error`) on `Main-HQ` via console `maintainer` login, and successfully resynchronized the entire operational configuration from `Backup-HQ`.
- **DHCP Reset Recovery:** Re-applied static interface parameters on `port1` following factory default fallback, restoring cluster WAN VIP reachability.
- **Windows Administrative Static Routes:** Added persistent static routes on the host PC (`route -p add 203.0.113.0 mask 255.255.255.248 192.168.85.201`) ensuring clean browser access to FortiGate Web GUIs.

9. Capstone Compliance Evaluation Matrix

Course Requirement	Target Specification	Status
Infrastructure Architecture	HQ (7 devices), BR1 (4 devices), BR2 (4 devices), Remote Client, ISP Core	**100% COMPLETED**
VLAN Segmentation	Users, Servers, Management VLANs with 802.1Q Trunks	**100% COMPLETED**
Inter-VLAN Routing	Centralized on FortiGate NGFW with zero-trust access control	**100% COMPLETED**
Dynamic DHCP	DHCP server scope enabled for User VLAN (.100 - .200)	**100% COMPLETED**
High Availability (HA)	Active-Passive cluster with dual heartbeats and verified failover	**100% COMPLETED**
Site-to-Site IPsec VPN	Hub-and-Spoke with Branch-to-Branch transit routing via HQ	**100% COMPLETED**
UTM Security Profiles	Antivirus, IPS, Web Filter, Application Control, SSL Inspection	**100% COMPLETED**
NAT & VIP Port Forwarding	Dynamic PAT on egress + Inbound VIP publishing (8080 → 80)	**100% COMPLETED**
Remote Access SSL-VPN	Tunnel mode with user authentication and destination resource ACL	**100% COMPLETED**
Server Hardening (Bonus)	Multi-tier micro-segmentation: Web (VLAN 20) → API (VLAN 25) → DB (VLAN 27)	**BONUS ACHIEVED**